

Interactive Finite-State Reactive Machine Assignment - An Interactive Digital Elevator

Using AI and an IDE of your choice, start a **completely new project** and design a program that will generate a **multi-floor elevator simulator** that uses an **finite-state machine (FSM)** model that demonstrates how a simple AI system behaves. A finite-state machine is a way of representing how a system moves between different states based on inputs or events. This assignment will help you understand how many reactive machine AIs and computer systems operate using rule-based logic.

In this assignment, you must create an **interactive, real-time elevator simulator** for an apartment building with **10 floors**. The simulator must allow the user to press the elevator buttons to determine which floors are requesting pickup and drop off. Remember, every floor needs to be able to request an up or down pickup, while also being able to select which floor they wish to travel to.

Your elevator simulator must include the following **states** at a minimum: Idle, MovingUp, MovingDown, DoorOpening, DoorClosing

Your elevator simulator must also demonstrate decision-making logic and queue handling, meaning that it should not pass floors requesting pickup while travelling... it must take the most optimum paths. It must also be prepared to handle edge cases like being idle with no buttons pressed, requesting the same floors, or having to respond when every single button is pressed.

Your code **must** be commented thoroughly. Machine-generated comments will not be sufficient.

This assignment will require the submission of an **AI DISCLOSURE FORM**. Any assignment submitted without the appropriate form will be given a zero grade.

Once your program is complete, you will be required to demonstrate your working simulator. Then, submit your project's code on LMS as a text file. Your AI Disclosure Form must be submitted as a PDF file.

Criteria	Exceptional	Meets Expectations	Satisfactory	Unsatisfactory
Functionality (Application) /10	The program functions exceptionally well, as intended, and is bug free.	The program functions well, as intended, and is bug free.	The program functions with some minor issues and bugs. Doesn't quite meet the requirements.	The program functions inconsistently and/or has numerous significant bugs.
Efficiency (Thinking/Inquiry) /10	The AI Disclosure Form is thoroughly complete. The strategy utilized was exceptionally logical and effective.	The AI Disclosure Form is complete. The strategy utilized was logical and effective.	The AI Disclosure Form is somewhat incomplete. The strategy utilized could have been refined.	The AI Disclosure Form is incomplete. The strategy utilized was illogical and ineffective.
Readability (Communication) /10	AI Disclosure Form is clear and concise. The code is exceptionally well organized and well commented.	AI Disclosure Form is relatively clear and concise. The code is organized and commented.	AI Disclosure Form lacks clarity. The code is difficult to follow and lacks effective comments.	AI Disclosure Form is low effort and lacks clarity. The code is difficult to follow or does not include comments.

